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RESEARCH-ARTICLE

MIC: Medical Image Classification Using Chest X-ray (COVID-19 & Pneumonia) Dataset with the Help of CNN and Customized CNN

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MIC: Medical Image Classification Using Chest X-ray (COVID-19 & Pneumonia) Dataset with the Help of CNN and Customized CNN

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Abstract

The COVID-19 pandemic has had a detrimental impact on the health and welfare of the world's population. An important strategy in the fight against COVID-19 is the effective screening of infected patients, with one of the primary screening methods involving radiological imaging with the use of chest X-rays. Which is why this study introduces a customized convolutional neural network (CCNN) for medical image classification. This study used a dataset of 6432 images named Chest X-ray (COVID-19 & Pneumonia), and images were preprocessed using techniques, including resizing, normalizing, and augmentation, to improve model training and performance. The proposed CCNN was compared with a convolutional neural network (CNN) and other models that used the same dataset. This research found that the Convolutional Neural Network (CCNN) achieved 95.62% validation accuracy and 0.1270 validation loss. This outperformed earlier models and studies using the same dataset. This result indicates that our models learn effectively from training data and adapt efficiently to new, unseen data. In essence, the current CCNN model achieves better medical image classification performance, which is why this CCNN model efficiently classifies medical images. Future research may extend the model's application to other medical imaging datasets and develop real-time offline medical image classification websites or apps.

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1 Introduction

A crucial stage in the analysis of medical images is medical image classification, which makes use of many elements to distinguish between distinct medical images, such as imaging modalities or clinical data. Clinicians may enhance the speed and accuracy of medical picture assessment through dependable medical image classification. Recent advancements in convolutional neural networks

(CNNs) have produced considerable advantages for the healthcare sector. The utilization of artificial intelligence-driven computer-aided diagnostic tools in medical settings has been extensively researched owing to these advancements. To get good diagnostic performance in medicine, CNNs can develop robust discriminative representations from extensive medical datasets. These algorithms yield results comparable to those of doctors and confirm their sufficient predictive capabilities [1-3].

Moreover, the emergence of deep learning in computer vision has enabled picture classification jobs to achieve unprecedented accuracy. A primary challenge in medical imaging related to inspection, diagnosis, and therapy is classification. Advancements in computing technology and methodology have enabled computer-aided solutions to potentially evaluate the clinical judgment of physicians. Several advanced deep convolutional neural network (CNN) models have been constructed and analyzed to categorize radiological images according to sickness or histological images. This requires detecting zones containing normal and cancerous cells [4-5].

However, various research have classified medical pictures using deep learning. One technique, which tackles difficulties with feature extraction and uncertainty quantification, uses a dynamic multi-scale convolutional neural network (DM-CNN) to improve medical picture classification. The dynamic multiscale feature fusion module (DMFF), hierarchical dynamic uncertainty quantification attention (HDUQ-Attention), multiscale fusion pooling method (MF pooling), and multiobjective loss (MO loss) are the four main components of the DM-CNN, which is a novel convolutional neural network architecture. Together, these sections establish the optimal convolution kernels for different feature map levels, modify attention weights, utilize Monte Carlo dropout to quantify uncertainty, and balance the trade-offs between classification accuracy and speed. During testing, the DM-CNN achieved state-of-the-art classification accuracy on four medical datasets: dermatology, histology, respiratory, and ophthalmology. It performed better than the other models. The model's capacity to measure uncertainty adequately and retain steady performance across several medical domains is crucial for clinical applications, as evidenced by the results. Despite using complex pooling approaches, the study concedes that it is restricted in its capacity to balance feature representation across various scales and may overfit. This means that additional refining is necessary to enhance these qualities while preserving the high accuracy and efficiency of the model [6].

Another method is to assess various fine-tuning procedures for applying pretrained convolutional neural networks (CNNs) to distinct medical imaging applications. They tested eight fine-tuning approaches—full fine-tuning, linear probing, and stepwise unfreezing—across three CNN architectures (ResNet-50, DenseNet-121, and VGG-19) and numerous medical domains (e.g., X-ray, MRI, and histology). The approaches were evaluated on several datasets, resulting in varying performance enhancements: some strategies enhanced accuracy by up to 11% in specific modalities. Compared with other topologies, DenseNet often benefits more distinctively from atypical fine-tuning. The results indicated the efficiency of adaptive fine-tuning approaches, particularly in settings where direct transfer learning falls short due to disparities between the source and medical image properties. However, the study's shortcomings include potential biases in dataset distribution and the computational expense involved with fine-tuning approaches, which might limit practical implementation in real-world circumstances where computational resources are constrained [7].

An other approach is HiFuse, a unique hierarchical multiscale feature fusion network created for medical picture classification. Adaptive hierarchical feature fusion blocks inverted residual multilayer perceptrons, global and local feature blocks, and other components are leveraged by HiFuse to include global and local features at multiple levels and boost classification accuracy. With classification accuracies of 85.85% and 86.12% on the ISIC2018 and Kvasir datasets, respectively, this technique helps to improve the semantic richness of the retrieved features and enhances accuracy across multiple medical datasets, including dermatological and histological datasets. The study suggests that, in spite of its excellent accuracy, it is tough to maintain balance in feature representation across different scales, which may result in overfitting of the model. Nonetheless, the model's performance maintains true across diverse datasets, confirming its usefulness in the classification of medical images [8].

As a result, only a small number of studies—like Ting et al. (2022)—have used the same dataset as this one. Their goal was to improve COVID-19-XR classification accuracy by implementing CNN architectures with more layers. Still, the model's accuracy was only 94% [9]. With a validation accuracy of 93.97% for multiclass instances, Patil and Narawade (2024) employed DL models to help with the diagnosis, treatment, and early identification of respiratory disorders such pneumonia and COVID-19 [10]. Elkamouny and Ghantous (2022) utilized eight deep learning models in addition to pretrained models for diagnosis; of these, the Inception-ResNet-v2 model outperformed the others with an accuracy of 95.3% [11]. In Patel S.'s evaluation of DenseNet (2021) for COVID-19 chest X-ray image classification, DenseNet201 was used, and the model's validation accuracy was 93.67% [12]. Even with this earlier research, there are still certain restrictions. Using the same datasets as earlier research, this paper presents a customized convolutional neural network (CCNN) and evaluates it against a standard CNN. The suggested CCNN performs better than rival models, according to the results.

The rest of the article is organized as follows: The study's methodology is thoroughly explained in Section 2, its findings are discussed in Section 3, and its conclusions are discussed in Section 4.

2 Methodology

2.1 Proposed Methodology

For this research purpose, CNN and customized CNN are appropriate. The diagram in Figure 1 shows the proposed methods of this study.

2.2 Dataset Collection

This study utilized Kaggle's Chest X-ray dataset, specifically focusing on cases with COVID-19 and pneumonia. The dataset comprises chest X-rays of individuals diagnosed with COVID-19, pneumonia, as well as those who are considered normal. The dataset is divided into two folders: "train" and "test". Every folder has three subordinate folders: "COVID-19", "PNEUMONIA", and "NORMAL". The collection comprises 6432 X-ray pictures, of which 20% are

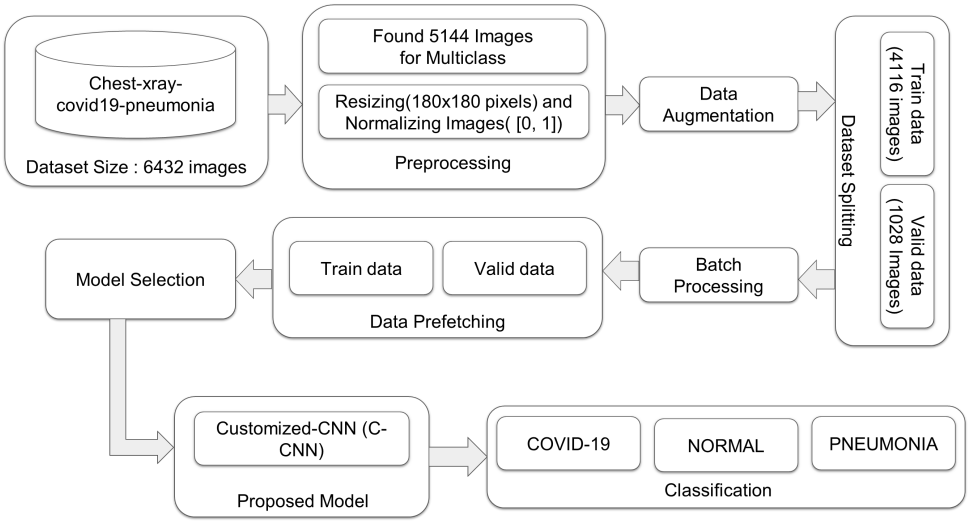

Figure 1: Proposed method of this current study.

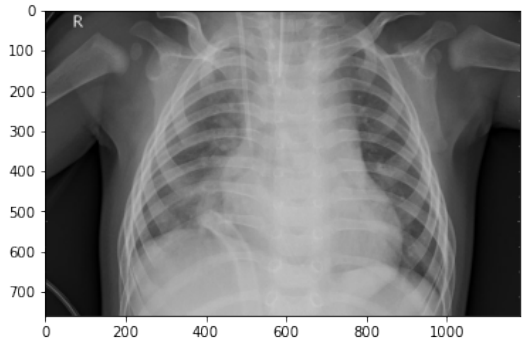

Figure 2: A pneumonia sample before preprocessing.

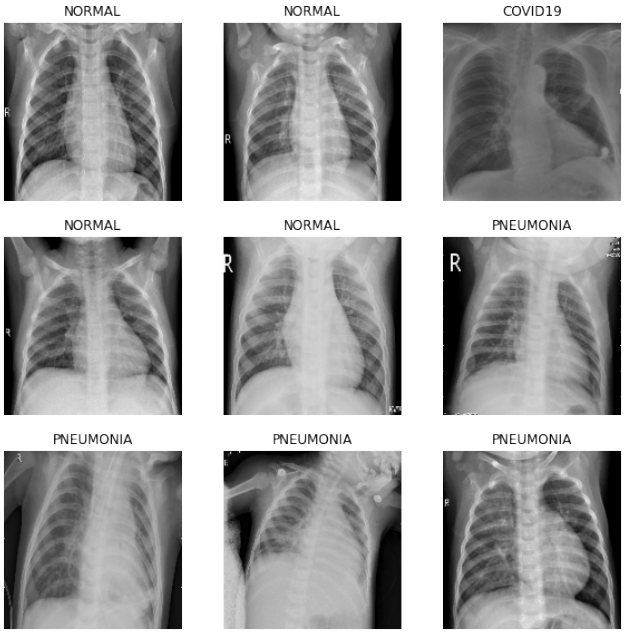

Figure 3: First 9 images after resizing and normalization of the dataset.

designated as test data [13]. Additionally, an image from the dataset is attached below, which is not preprocessed.

2.2.1 Data Preprocessing. For this study, data preprocessing is a critical step in building effective machine learning models, especially when dealing with image data such as medical images. There are several common data preprocessing techniques used in the context of medical image analysis, as illustrated below. However, figure 2 shows A pneumonia sample before preprocessing.

2.2.2 Resizing and Normalizing Images. For the present study, images were resized to a consistent shape (180x180 pixels) to ensure uniformity across all inputs. This is crucial for the model to process them efficiently. Normalization is achieved by rescaling the values of each pixel to fit within the range of 0 to 1. This helps the acceleration of convergence during training by assuring that all features (pixel values) participate equally [14]. Moreover, figure 3 shows the first images after resizing and normalization of the dataset.

2.2.3 Data Augmentation. The present study applied a data augmentation approach to artificially increase the dataset's size by generating modified copies of the images inside the dataset. Additionally, images are subjected to random horizontal flipping in order to replicate various orientations. Images are randomly rotated by a small angle (5% of 360 degrees). Random Zoom: The images are zoomed in or out by a certain percentage (10%). However, augmentation helps in building robust models by introducing variability in the training data, which can improve generalization [15]. Additionally, Figure 4 shows the first 9 images after data augmentation.

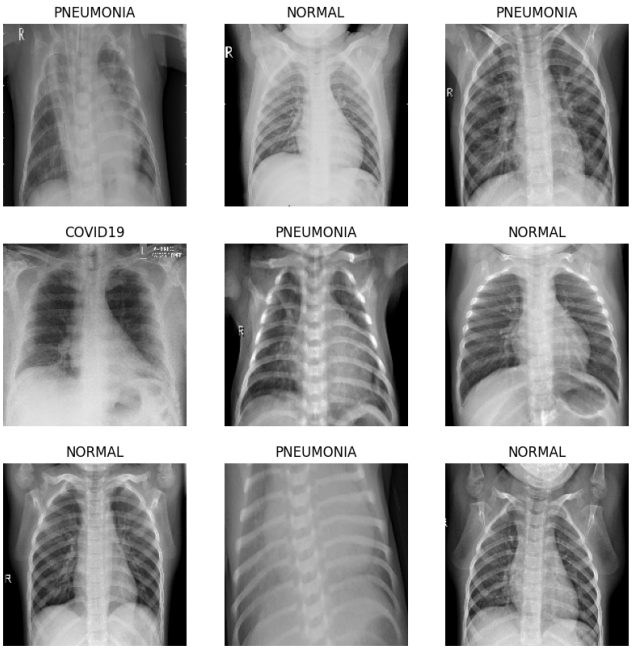

Figure 4: First 9 images after data augmentation of the dataset.

2.2.4 Data Splitting. The dataset is often divided into training and validation sets for this kind of investigation. This allows the model to be trained on one set of data, refined on another, and finally tested on data that hasn't been seen yet. Usually, 20% of the training is utilized for validation, which allows us to track the model's performance as it is being trained. A 20% training dataset was also employed in this investigation.

2.2.5 Batch processing. The data used in the present study were processed in batches (32 images per batch). This is a practical approach for making efficient use of memory and improving computational speed. Batching also allows the model to update its weights after seeing several examples and tends to make the learning process more stable.

2.2.6 Prefetching. This study used the prefetching technique to preload data into buffer memory to speed up the training process. This technique ensures that the next batch of data is readily available for the model to process as soon as it finishes with the current batch, reducing the time spent waiting for data loading.

2.3 Model Selection

For this current study selected two models, one is CNN and another one is CCNN which are described below.

2.3.1 CNN. The CNN model's construction includes numerous layers that successively process and simplify the image input. Initially, an image augmentation phase enriches the input data to help the model generalize better from the training data. Following this, the network includes numerous layers that apply filters and minimize the quantity of the data, interspersed with dropout layers that randomly disregard sections of the data during training to avoid the model from memorizing the training data too closely. The network complexity rises by adding more filters at deeper levels. After processing through these layers, the data are flattened and put to a thick layer for final classification. The configuration of the output layer depends on whether the model is divided into two categories (using sigmoid activation) or more (using softmax activation). On the other hand, the model incorporates learning rate scheduling, which adjusts the model's learning process as training continues, and early stopping to prevent overfitting. The model learns to classify new images using the patterns it has learnt during training, which is carried out using specified epochs. Additionally, the CNN model architecture's pseudocode is supplied below.

Algorithm 1 Pseudocode of CNN model architecture

```
function MAKE_MODEL(input_shape, num_classes):
    inputs ← keras.Input(shape=input_shape)
      x ← data_augmentation(inputs)
      x ← layers.experimental.preprocessing.Rescaling(1.0/255)(x)
      x ← apply several convolutional, activation, and pooling
layers
      x ← layers.Flatten()(x)
      x ← layers.Dense(256, activation='relu')(x)
      if num_classes = 2 then
        activation ← "sigmoid"
          units ← 1
      else
          activation ← "softmax"
          units ← num_classes
       x ← layers.Dropout(0.5)(x)
      outputs ← layers.Dense(units, activation=activation)(x)
      return keras.Model(inputs, outputs)
  end function
```

2.3.2 CCNN. To classify images, a CCNN model is developed with the help of the Keras package. The function 'make_model' defines the model and requires two parameters, 'input_shape} and 'num_classes}. Without acquiring new data, it begins by taking in input photos and applying data augmentation to expand the variety of training examples. Next, the resampled images are assigned values ranging from 0 to 1, therefore enhancing the training efficiency.

The model's core comprises of multiple convolutional layers, each of which is isolated from the others by batch normalization, activation of rectified linear units (ReLUs), max pooling to minimize dimensionality, and dropout to reduce overfitting. The architecture is clearly tweaked, with increased dropout rates and varied filter sizes applied to different layers in an effort to enhance speed.

The global average pooling layer, which comes after the convolutional layers in the model, contributes in lowering the overall number of parameters and making the model simpler. To further prevent overfitting, a dense layer with ReLU activation is added, and then there is another dropout, but this time at a rate of 0.5.

The model's output layer varies according to the number of classes: a sigmoid activation is used for multiclass classification, while a softmax activation is used for binary classification (num_classes == 2}). Because of its adaptability, the model can change depending on the classification task.

The Adam optimizer, which uses a modest learning rate to guarantee gradual and steady learning, is used to generate the model. It makes use of a sparse categorical cross-entropy loss function. To improve efficiency, the training technique includes callbacks, such as early stopping, to terminate training if the validation loss does not grow. Furthermore, a learning rate scheduler with adjustable decay rates is used to progressively lower the learning rate following a predetermined number of epochs.Basically, this model is tailored to perform better and more efficiently for image classification tasks by making specific changes to its architecture and training plan. It is evident from the highlighted setup and customization locations where certain changes have been made for possible improvement [16]. Furthermore, the CCNN is intended to improve the categorization of medical images, specifically for the identification of pneumonia and COVID-19 from chest X-ray images. Advanced data augmentation methods that increase the diversity of training instances and boost the generalization of the model are used by the CCNN. These methods include random horizontal flipping, small-angle rotations, and zooming. Rescaling images to values between 0 and 1 increases the effectiveness of the training process. Multiple convolutional layers with batch normalization and ReLU activation make up the core of the CCNN. Filter sizes of 32, 64, 128, and 256 are used to capture varying levels of information. Dropout layers with rates rising from 0.3 to 0.5 stop overfitting, whereas max pooling layers lower dimensionality and computing burden. A global average pooling layer minimizes the overall number of parameters, simplifying the model and lowering the danger of overfitting. The output layer uses sigmoid or softmax activation to adjust to binary or multi-class classification tasks, while the dense layer with 256 units and ReLU activation processes the flattened feature maps. The CCNN performs better as a result of these architectural decisions since they allow it to extract strong features, reduce overfitting, and adapt effectively to various classification tasks. With a validation accuracy of 95.62% and a validation loss of 0.1270, the CCNN has outperformed other models in terms of learning from training data and reliably classifying newly discovered data. Additionally, the CCNN model architecture's pseudocode is provided below.

Algorithm 2 Pseudocode of CCNN model architecture

```
function MAKE_MODEL(input_shape, num_classes):
    inputs ← keras.Input(shape=input_shape)
    x ← data_augmentation(inputs)
    x ← layers.experimental.preprocessing.Rescaling(1.0/255)(x)
    # Adding more convolutional layers and adjusting filter sizes
    for filters in [32, 64, 128, 256]:
        x ← layers.Conv2D(filters, 3, padding="same",
kernel_regularizer=l2(0.01))(x)
        x ← layers.BatchNormalization()(x)
        x ← layers.Activation("relu")(x)
        x ← layers.MaxPooling2D()(x)
        x ← layers.Dropout(0.3)(x)  # Increased dropout rate
    x ← layers.GlobalAveragePooling2D()(x)  # Using
GlobalAveragePooling2D before the dense layer
    x ← layers.Dense(256, activation='relu',
kernel_regularizer=l2(0.01))(x)
    x ← layers.Dropout(0.5)(x)
    if num_classes = 2 then
        activation ← "sigmoid"
        units ← 1
    else
        activation ← "softmax"
  units ← num_classes
    outputs ← layers.Dense(units, activation=activation)(x)
    return keras.Model(inputs, outputs)
end function
```

Figure 5: Training and validation loss of CNN.

3 Result

3.1 CNN training and validation loss

Figure 5 shows the training and validation loss of a convolutional neural network (CNN) over the first 24 epochs, which is part of a training regimen with a total of 50 epochs. Initially, both the training and validation losses are high, starting at approximately 0.89. Rapid improvement is observed by the 5th epoch, with the training loss significantly reduced to approximately 0.27 and the validation loss to approximately 0.17. However, there is a notable spike in the validation loss to 0.51 by the 8th epoch, suggesting potential overfitting at that stage. After this spike, the losses gradually stabilize, with the training loss being consistently lower than the validation loss. By the 20 h epoch, the training loss has further decreased to approximately 0.14, indicating effective learning and fitting to the training data, while the validation loss has also decreased and leveled off, reflecting better generalization to new data.

3.2 CNN training and validation accuracy

The first 20 epochs of CNN model training and validation accuracy are shown in the figure 6. The training accuracy starts at 63.22% and improves quickly, reaching above 95.53% by the 24th epoch. The validation accuracy begins at 66.34% and follows a similar upward trend but exhibits more volatility, which is particularly noticeable

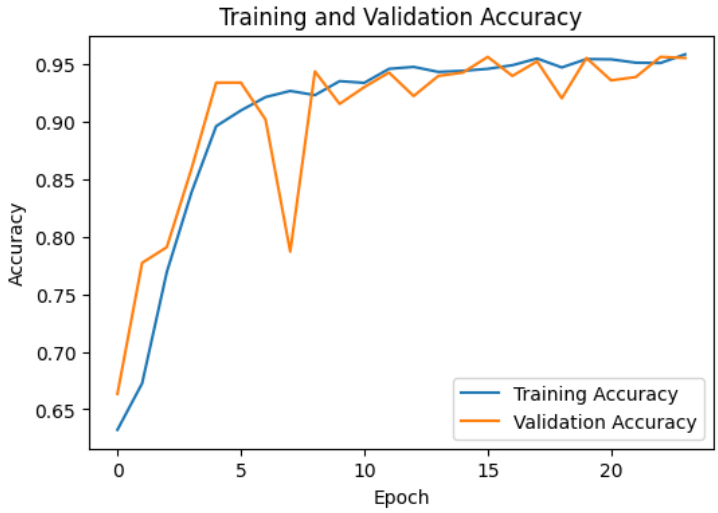

Figure 6: Training and validation accuracy of CNN.

around the 6th and 8th epochs, where it dips before rising again. This fluctuation might indicate moments when the model struggled with new data before adapting. By the 24th epoch, both accuracies are high, with 95.53% validation accuracy, which is slightly below the training accuracy of 95.85%, suggesting that while the model is performing well, there is still room for improvement.

3.3 CCNN training and validation loss

Figure 7 shows the training and validation loss of the C-CNN model over 24 epochs. Both losses start high (over 11) and decrease steadily, indicating that the model is learning from the data. The losses closely follow each other, which suggests good generalizability without significant overfitting. However, there is a notable increase in the validation loss at the 20th epoch, which might indicate that the model is starting to overfit. By the 24th epoch, both losses decreased significantly, with the training loss being slightly lower than the validation loss. This shows that the model's performance has greatly improved throughout the training process.

3.4 C-CNN Training and Validation Accuracy

Figure 8 presents the training and validation accuracy of a convolutional neural network model over 24 epochs. The training accuracy starts from a lower value but rapidly increases, stabilizing at approximately 90% to 95% for the majority of the training. This indicates that the model consistently performs well on the training data throughout the later epochs. In contrast, the validation accuracy decreases but significantly improves as the number of epochs increases, suggesting that the model gradually learns to generalize better to unseen data. However, the validation accuracy shows considerable variability, particularly a sharp decline around the 20th epoch, which suggests potential issues such as overfitting or an anomaly in the validation data at that point. Despite this, both accuracies reach high values by the final epochs (training accuracy 95.70% and validation accuracy 95.62%), demonstrating overall successful learning and adaptation by the model.

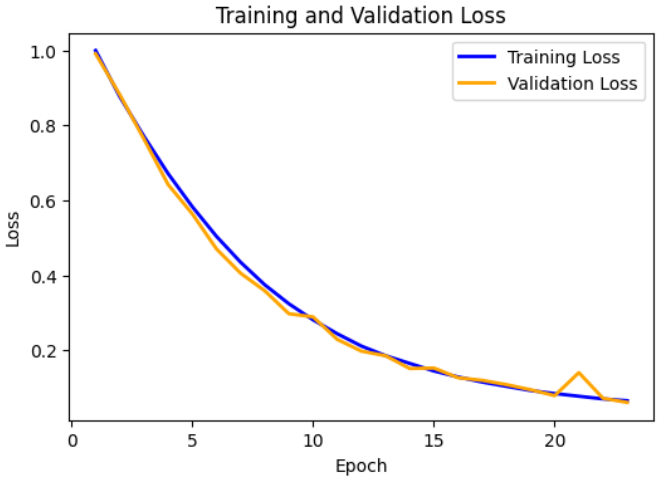

Figure 7: Training and validation loss of CCNN.

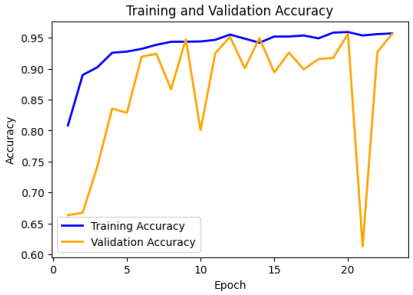

Figure 8: Training and validation loss of CCNN.

Table 1: Comparison between previous and current studies

Reference	Model	Validation Accuracy	Validation Loss
[10] Patil and Narawade (2024)	Deep CNN	0.9397	0.3719
[9] Ting et al.(2022)	CNN architectures with additional layers	0.94	Not provided
[11] Elkamouny and Ghantous (2022)	Inception-ResNet-v2	0.953	Not provided
[12] Patel, S. (2021)	DenseNet201	0.9367	0.1653
Proposed	Customized-CNN (C-CNN)	0.9562	0.1270

3.5 Comparison

Table I presents a comparison between prior studies and the current study. According to the data, the suggested model in this study showed better results compared to the other models in terms of validation accuracy. The validation loss is a metric that quantifies the performance of a model on the validation set. It is calculated by applying the model's loss function to the validation data and averaging the results. Lower validation loss values indicate that the model is predicting the validation data more accurately.

4 Conclusion

This work represents a notable advancement in the field of medical image classification. The performance of the customized convolutional neural network (CCNN) developed herein is validated with an accuracy level of 95.62%. This CCNN model outperforms the CNN model and other existing studies that used the same dataset. This study used rigorous data preprocessing steps (resizing, normalizing, and augmenting), training and tuning strategies to solve the challenges of medical image classification. The customized model could still be optimized by expanding to other larger medical imaging datasets and more diverse datasets. This study contributes valuable insights to the academic field. This study also contributes to significant improvements in healthcare and helps to efficiently classify medical images.

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